

Math 170E

Week 8

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1 Ch4

Example 1. (PSI example 4.1.3) The joint pmf of $X&Y$ is

$$f(x, y) = c(x + y) \text{ for } x = 1, 2, y = 1, 2, 3$$

1. Find c . (A) 6 (B) $\frac{1}{6}$ (C) 12 (D) 21 (E) $\frac{1}{21}$

2. Find marginal pmf $f_X(x)$. (A) $\frac{2x+y}{9}$ (B) $\frac{2x+1}{7}$ (C) $\frac{x+1}{7}$ (D) $\frac{x+2}{7}$ (E) $\frac{x+3}{9}$

3. Find marginal pmf $f_Y(y)$. (A) $\frac{2y+3}{21}$ (B) $\frac{y+6}{21}$ (C) $\frac{y+1}{7}$ (D) $\frac{2y+1}{14}$ (E) $\frac{3y+1}{21}$

4. Find $P(X > Y)$ (A) $\frac{1}{21}$ (B) $\frac{2}{21}$ (C) $\frac{1}{7}$ (D) $\frac{5}{21}$ (E) $\frac{2}{7}$

5. Find covariance of $X&Y$. Are they independent?

6. Find correlation coefficient of $X&Y$, i.e. ρ .

7. Find the conditional probability distribution $g(x|y)$.

Example 2. (Old Midterm Problem) Assume that the high school grade point average of a student, X , and the college freshman gpa of a student, Y , are distributed according to a bivariate normal distribution with parameters $\mu_X = 3$, $\mu_Y = 4.3$, $\sigma_X = 0.4$, $\sigma_Y = 1.25$, and $E[XY] = 13.3$

- Find the correlation coefficient ρ of X and Y .
(A)0.2 (B)0.25 (C)0.4 (D)0.6 (E)0.8
- Are X and Y independent random variables? Why or why not?
- Compute the probability $P(3.4 < X < 4.2)$ (A)0.1574 (B)0.2215 (C)0.324 (D)0.4512 (E)0.52
- Compute the probability $P(7.8 < Y < 8.55 \mid X = 5)$

HINT: Mean and variance of bivariate normal Y conditioned on $X = x$

$$\mu_{Y|x} = \mu_Y + \rho \frac{\sigma_Y}{\sigma_X} (x - \mu_X);$$

$$\sigma_{Y|x}^2 = \sigma_Y^2 (1 - \rho^2).$$

- (A)0.136 (B)0.211 (C)0.261 (D)0.36 (E)0.611

Example 3. (PSI 4.3-10) Let $f_X(x) = 1/10$, for $x = 0, 1, 2, \dots, 9$, and $h(y|x) = 1/(10 - x)$, for $y = x, x + 1, \dots, 9$. Find

- $f(x, y)$
- $f_Y(y)$.
- $E(Y|x)$.

Example 4. Let $X \sim \text{Unif}(2, 4)$ ¹ and let $Y \sim \text{Pois}(X)$ ². Using *Laws of Total Expectation and Variance*, compute $\mathbb{E}Y$ and $\text{Var}(Y)$.

¹Meaning X has distribution uniform distribution $\text{Unif}(2, 4)$

²More precisely, this means $Y \mid X = x$ (Y given $X = x$) has distribution $\text{Pois}(x)$

Example 5. (PSI 4.3-5) Let X and Y have a trinomial distribution with $n = 2$, $p_X = 1/4$, and $p_Y = 1/2$. Find $E(Y|x)$.